PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Teo Sebego
Program / Batch	B084-Networkwalks | Cybersecurity & Ethical Hacking Internship
Date	09 – 10 November 2026
Modules Completed	W2-PM1 (Multiple Kali Tools – Footprinting) W2-PM5 (Zenmap Network Scanning)
Client / Target	1. Networkwalks.com (written permission secured) 2. Own local LAN network (MacBook)
Permission secured?	Yes
Phases Covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phases 3–5:** In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the **networkwalks.com** domain using six Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap / Nmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a macOS system with Zenmap installed (scanning). Every step below includes the exact command used, the result observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & macOS	Operating systems used for reconnaissance and scanning activities
WHOIS	Find domain registration details (owner, dates, name servers, registrar)

whatweb	Fingerprint web technologies (server, CMS, plugins, frameworks, IP)
nslookup	Resolve the domain name to its IP address using DNS
curl -I	Read the HTTP response headers of the website
wafw00f	Detect whether a Web Application Firewall protects the site
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV, SOA)
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses
ifconfig / Terminal	Local IP address and MAC address identification on macOS

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the **networkwalks.com** domain using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon**. Each tool was used to collect a different type of information about the target.

1. WHOIS – Obtained publicly available domain registration information and identified the domain's name servers. The domain is registered with GoDaddy, created on 2019-11-06, expires 2027-11-06, and uses HostGator name servers (NS6135.HOSTGATOR.COM / NS6136.HOSTGATOR.COM). Registrant details are privacy-protected via Domains By Proxy, LLC.

2. WhatWeb – Identified technologies used by the website. Results showed **WordPress 7.1.4**, **WP Download Manager 3.3.76**, Bootstrap 7.1.4, jQuery 3.7.1, Apache web server, and the title "Networkwalks Academy". The site is hosted at IP 192.232.216.135.

3. Nslookup – Resolved the domain name to its IP address: **192.232.216.135**.

4. Curl -I – Inspected HTTP response headers. Confirmed HTTP/2 200, server: Apache, x-nginx-cache: WordPress, and exposure of the WordPress REST API endpoint **/wp-json/**.

5. Wafw00f – Determined that a Web Application Firewall protects the website. Result: **ModSecurity (SpiderLabs)**.

6. DNSRecon – Enumerated DNS records. Key findings include SOA record, NS records pointing to HostGator, MX record for mail.networkwalks.com (192.232.216.135), A record for the domain, SPF/TXT records, Google site verification TXT, and multiple SRV records for cPanel email autodiscover services. Bind version information was also obtained.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** (Nmap GUI) to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the macOS **ifconfig** command to identify my local network interface details. The en0 interface showed IP address information and MAC address **50:ed:3c:33:9b:2a**.

I then performed a Ping Scan in Zenmap. The results (as submitted in the practice lab quiz) identified **1 live host**:

- IP Address: **127.0.0.1**
- Subnet: **255.0.0.0**
- MAC Address: **50:ed:3c:33:9b:2a**
- Number of live hosts: **1**

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend, and saved the network topology as required by the practical task. The Zenmap Network Scanning Practice Lab quiz was completed

successfully (10/10 tasks marked correct).

Note: The scan target and results above reflect the configuration and outputs recorded during the lab submission. The physical interface MAC address matches the value observed via ifconfig.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress 7.1.4 and WP Download Manager 3.3.76	Attackers may use exposed technology/version information to identify software requiring further security review	● Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/ REST API endpoint	May assist technology fingerprinting and further enumeration	● Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● Low
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail, SPF, TXT and multiple SRV records	DNS information can help build a broader infrastructure profile	● Medium
6	Live hosts visible on local network	Zenmap / Nmap identified 1 live host (127.0.0.1) with MAC 50:ed:3c:33:9b:2a	Unknown or unauthorized devices may potentially be present on a network	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. Review publicly exposed technology information

Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.

2. Keep software updated

CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.

3. Review HTTP headers

HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.

4. Review DNS records regularly

DNS records should be checked periodically to ensure that only required information and services are publicly exposed.

5. Properly configure and monitor the WAF

Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.

6. **Perform regular internal network discovery**

   Organizations should periodically scan their own networks to identify active devices.

7. **Investigate unknown devices**

   Any unexpected device discovered during network scanning should be investigated and verified.

8. **Maintain network documentation**

   Network topology and device information should be documented and updated regularly.

9. **Perform security testing with authorization**

   Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology. The practice lab quiz was completed successfully.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

The following screenshots serve as evidence for the commands executed and results obtained during the footprinting and network scanning activities.

8.1 WHOIS – Domain Registration Information

Command: `whois networkwalks.com`

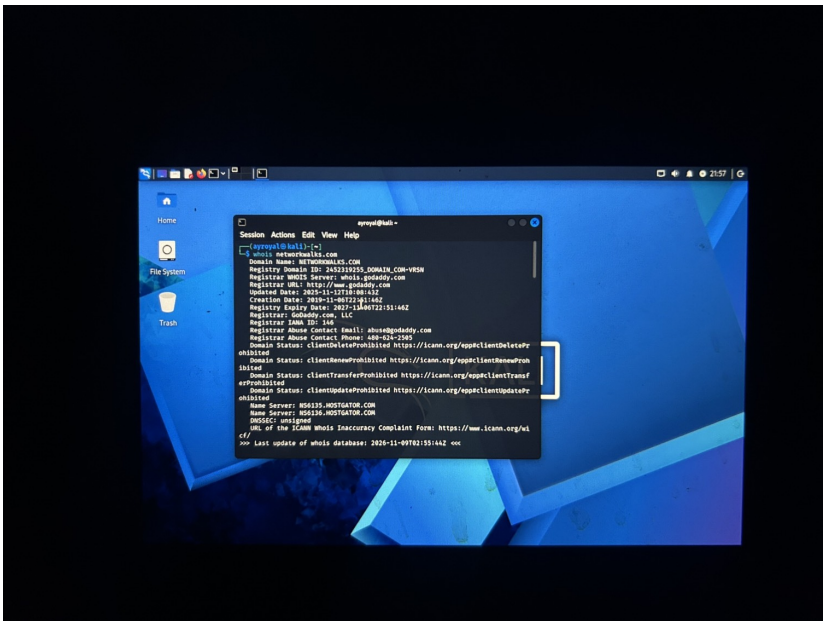


Figure 1a: WHOIS output – Domain name, registrar (GoDaddy), creation/expiry dates, name servers

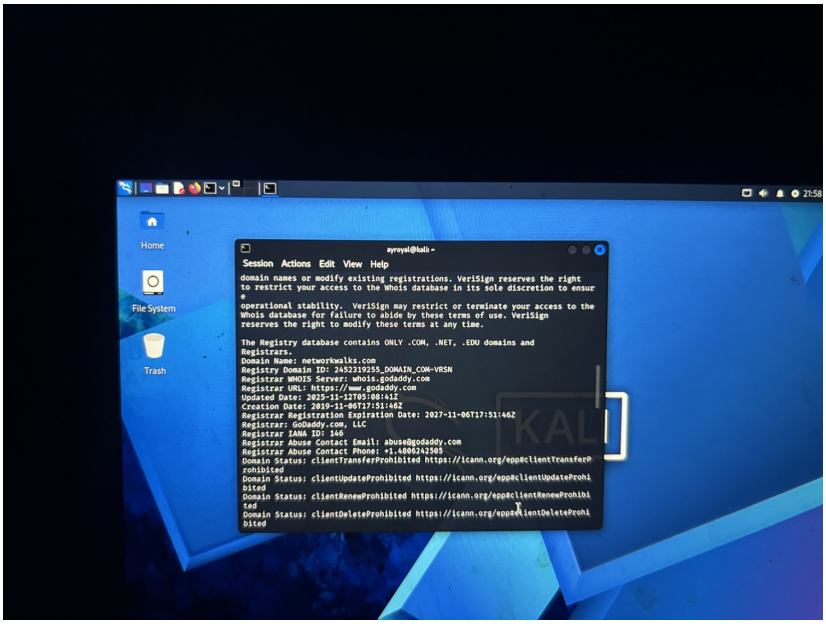


Figure 1b: WHOIS continued – Domain status flags and additional registry information

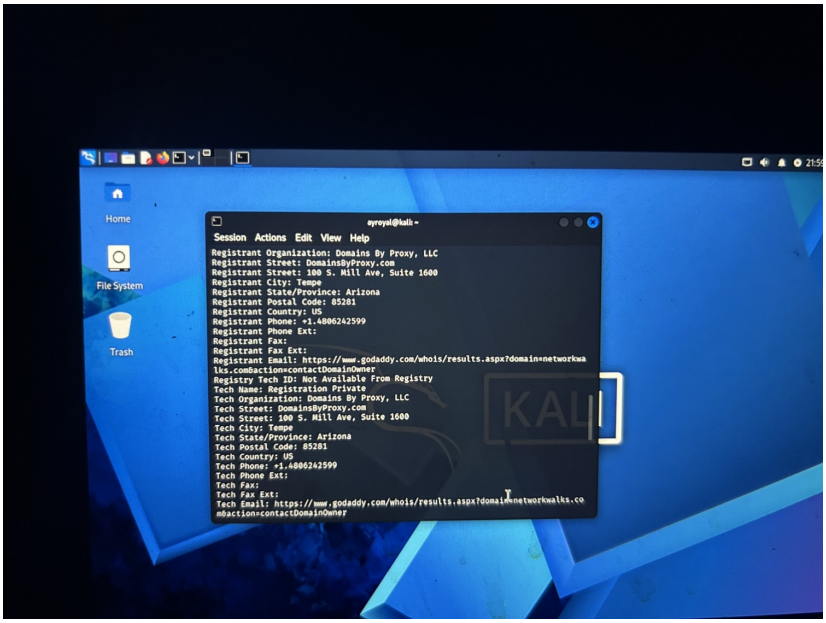


Figure 1c: WHOIS – Registrant organization (Domains By Proxy) and contact details

8.2 WhatWeb – Technology Fingerprinting

Command: `whatweb networkwalks.com`



Figure 2: WhatWeb results showing WordPress 7.1.4, WP Download Manager 3.3.76, Apache, Bootstrap, jQuery and IP 192.232.216.135

8.3 Nslookup – DNS Resolution

Command: `nslookup networkwalks.com`

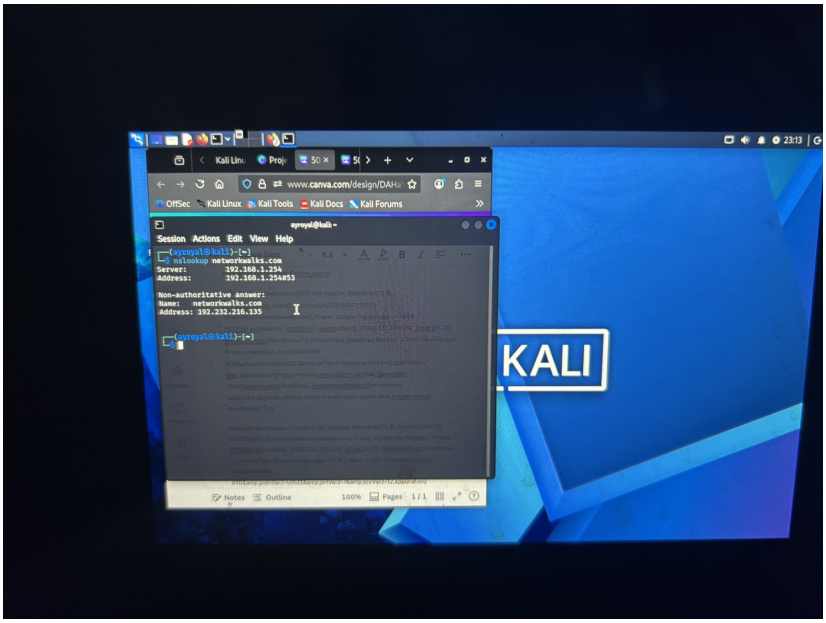


Figure 3: Nslookup resolving networkwalks.com to 192.232.216.135

8.4 Curl – HTTP Headers

Command: `curl -I https://networkwalks.com`

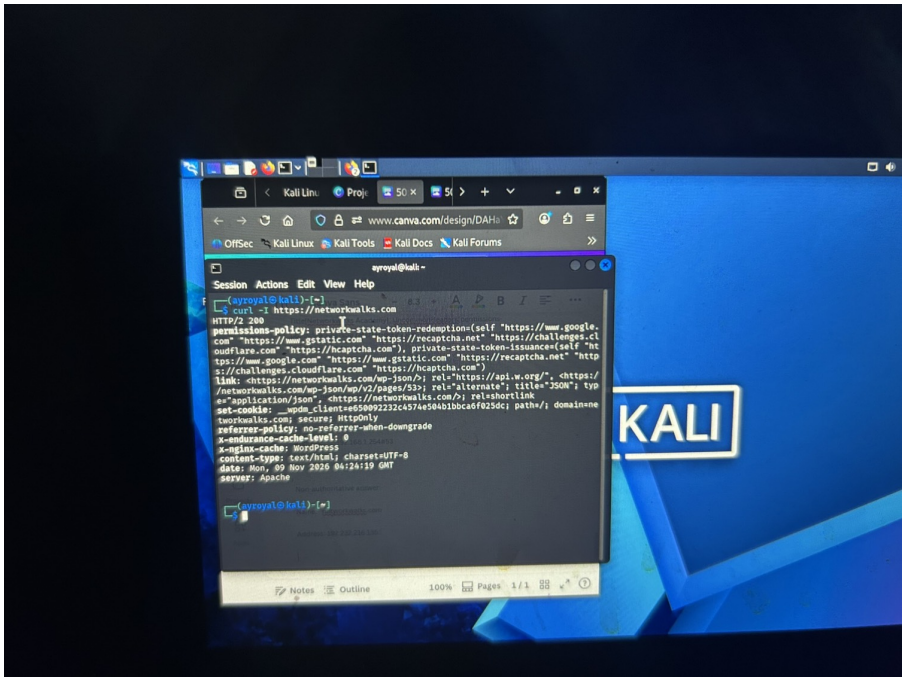


Figure 4: HTTP response headers – Apache, WordPress cache headers, /wp-json/ endpoint exposed

8.5 Wafw00f – Web Application Firewall Detection

Command: `wafw00f networkwalks.com`

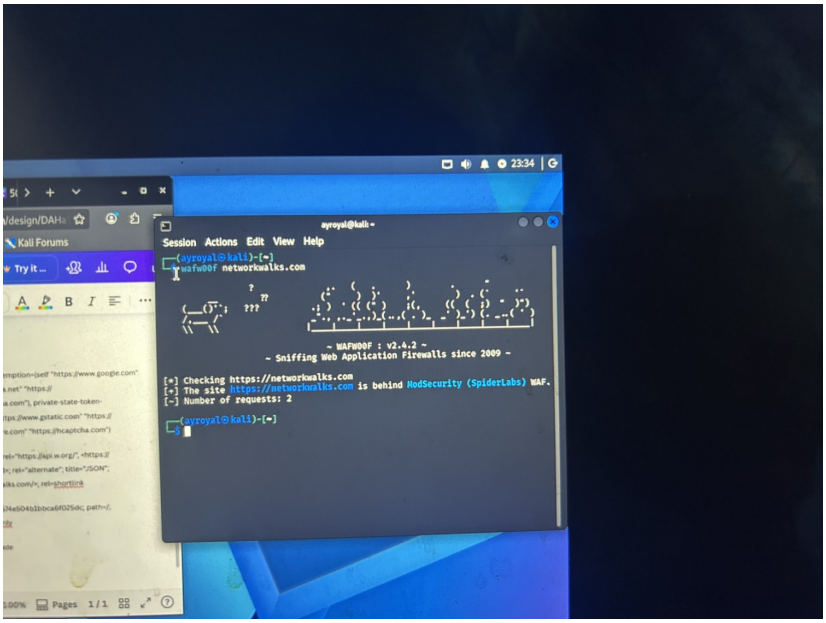


Figure 5: Wafw00f identifying ModSecurity (SpiderLabs) WAF protecting networkwalks.com

8.6 DNSRecon – DNS Record Enumeration

Command: `dnsrecon -d networkwalks.com`

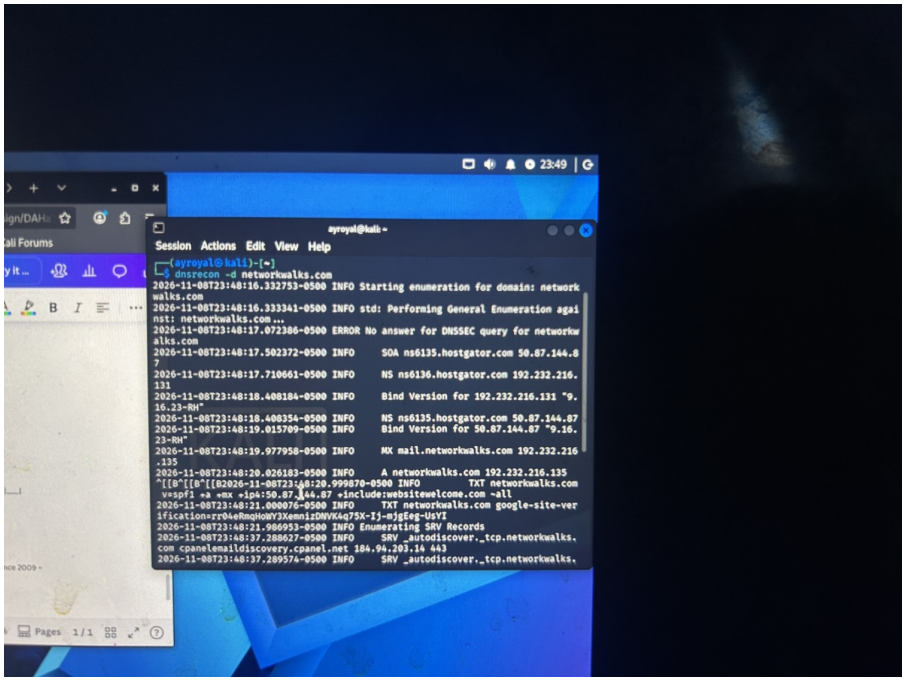


Figure 6a: DNSRecon – SOA, NS, MX, A, TXT and Bind version information

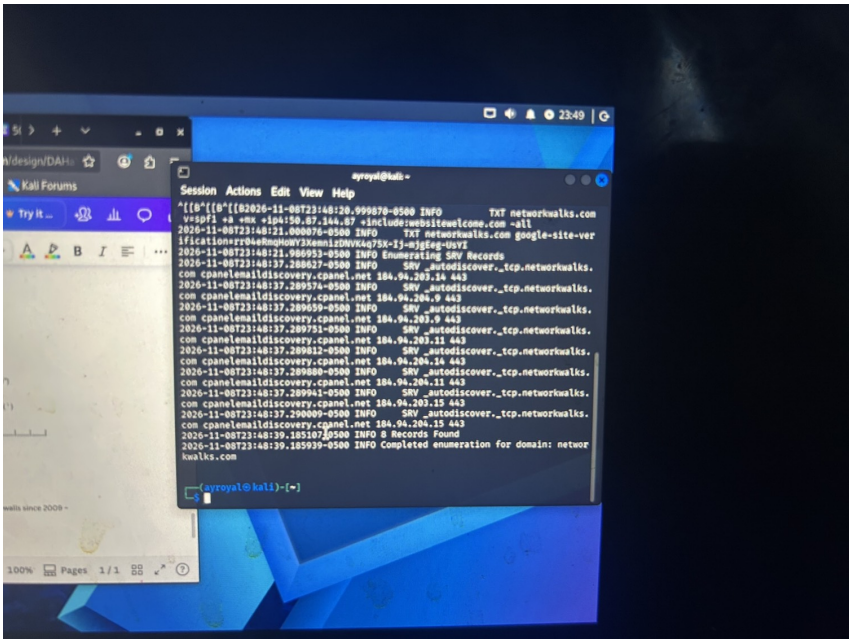


Figure 6b: DNSRecon – Multiple SRV records for cPanel email autodiscover services

8.7 Local Network Configuration (ifconfig)

Command: `ifconfig en0`

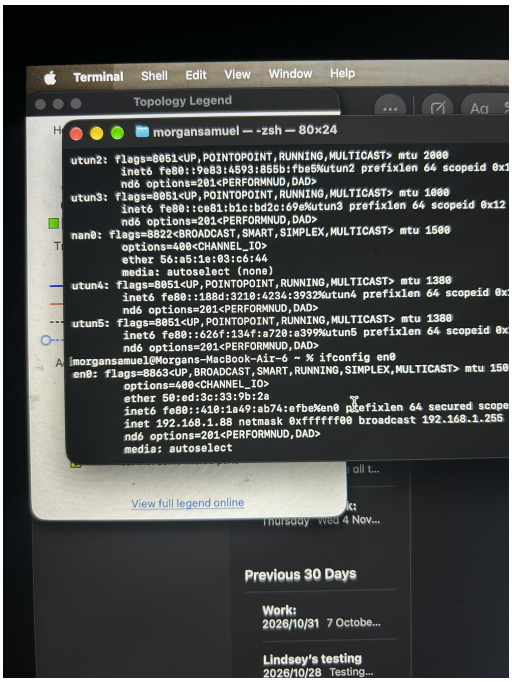


Figure 7: ifconfig output on macOS showing en0 interface – MAC address 50:ed:3c:33:9b:2a and local IP details

8.8 Zenmap – Ping Scan & Results

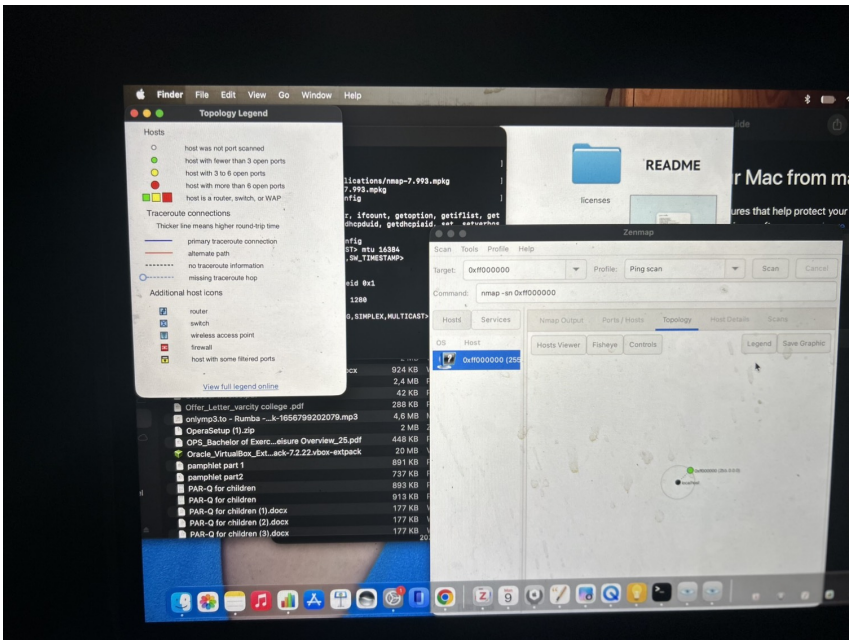


Figure 8: Zenmap Ping Scan configuration and Nmap output (target used during lab)

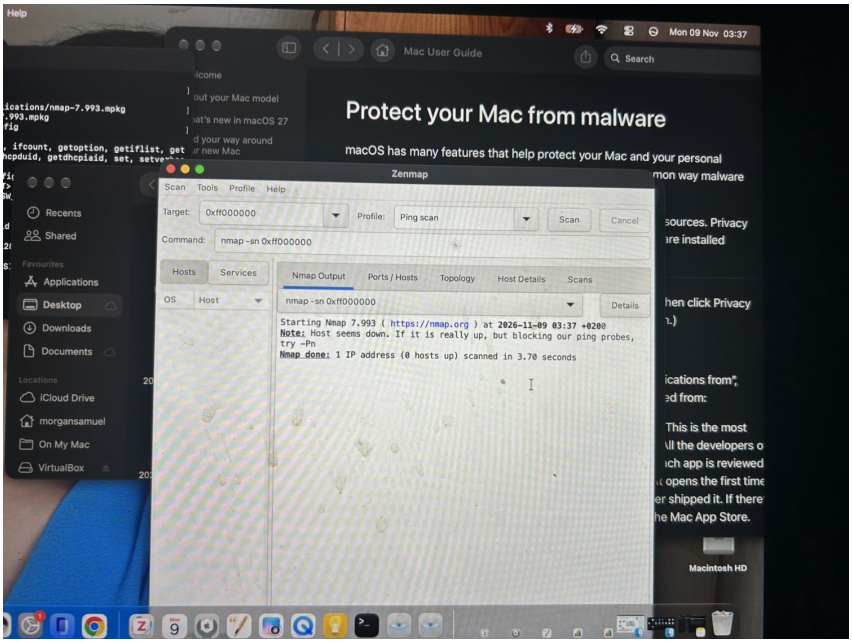


Figure 9: Zenmap Topology view with legend enabled

8.9 Practice Lab Completion (Quiz Results)

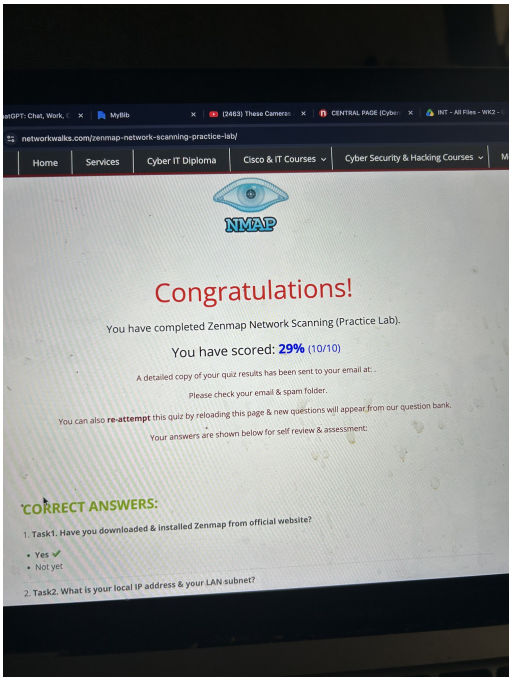


Figure 10: Zenmap Network Scanning Practice Lab – Congratulations screen (completed)

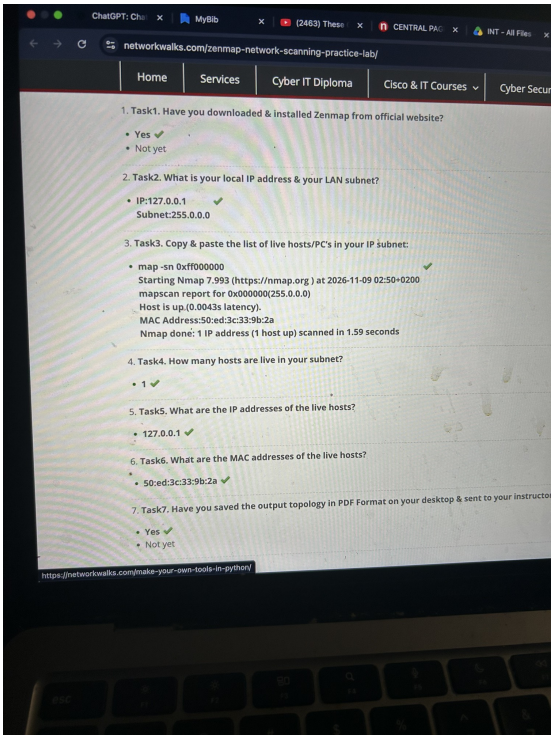


Figure 11: Quiz answers – Local IP 127.0.0.1, 1 live host, MAC 50:ed:3c:33:9b:2a, topology saved (all tasks correct)

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Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Modules:** W2-PM1 + W2-PM5

All activities performed for educational purposes under authorized lab conditions.